

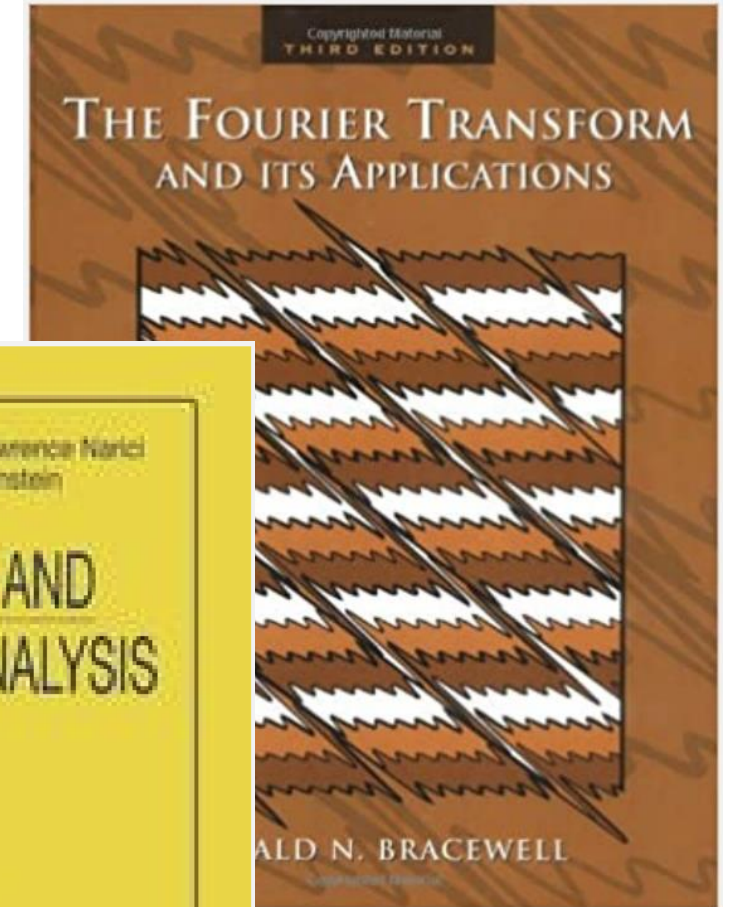
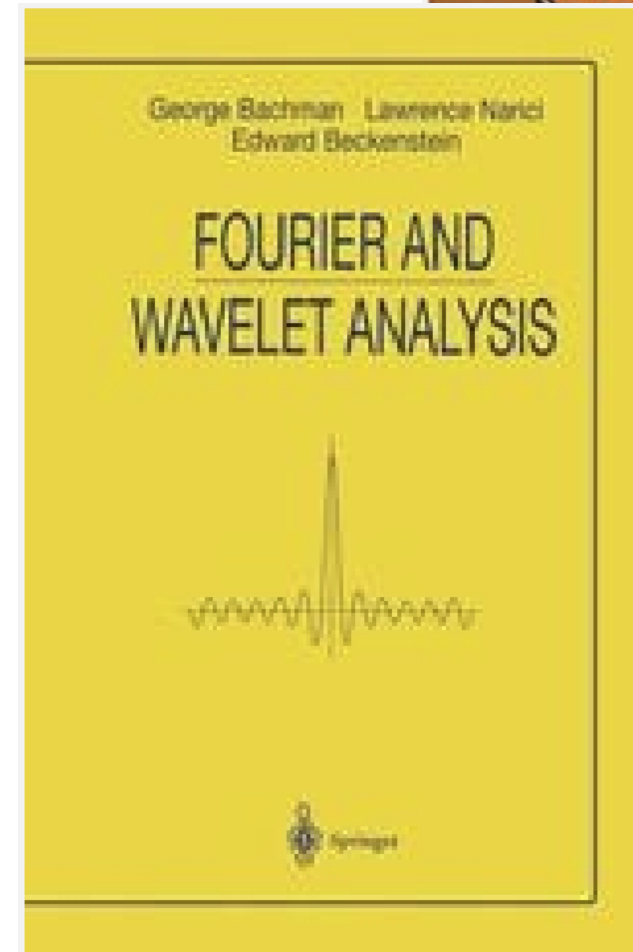
Multidimensional Fourier Transform

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Agenda

- Overview of the general idea behind Fourier series
- Fourier transform and basic properties
- Extension of the FT to multiple dimensions



Signals, systems, and measurements

- We want to study a particular system, by measuring a signal f .
- *How is the signal measured in practice?*
- **Possible solutions:**
 - We have an instrument I , which can perfectly measure the signal in time, $f(t)$.
 - Alternatively (and practically), we may have a number of samples of the signal, e.g., at equidistant sampling intervals, $f[n] = f(nT)$, where T is the sampling period.

The Fourier Series

- Recall that we want to measure a real signal.
- Consider an orthogonal basis of functions, f_n .
- Using this, we may express our signal as a linear combination of these functions in a bounded interval, $t \in [-T/2; T/2]$:

$$f(t) = \sum_{n=-\infty}^{\infty} a_n f_n(t)$$



Coefficient associated with the n'th basis function

The Fourier Series

- Already in the 18th century, Fourier claimed that any **periodic** function can be written as a series of **trigonometric** function.
- Mathematically, this can be expressed as:

$$f(t) = \sum_{k=-\infty}^{\infty} a_k e^{j2\pi kt/T}$$

- For analysis, we can use the inverse relationship:

$$a_k = \frac{1}{T} \int_T f(t) e^{-j2\pi kt/T} dt$$



Importance of Periodic Signals

- Many physical phenomena exhibit periodic behaviour and can thus be modeled and analyzed using periodic basis functions:
 - Sound waves (musical instruments).
 - AC electrical signals
 - Planetary motion (orbits)
 - Pendulum motion (simple harmonic oscillator)
 - Rotating machinery
 - Heartbeats (ECG)
 - Seasonal climate patterns.
 - Biological rhythms (circadian rhythms)
- Moreover, sinusoidal functions are eigenfunctions of LSI systems.

Beyond Periodic Functions

- While many natural phenomena are periodic, not all are.
- Moreover, even some of the aforementioned phenomena are not perfectly periodic.
- We can extend the Fourier series, by
 - considering an unbounded interval, $T \rightarrow \infty$.
 - considering the frequency as being continuous.
- This functional operator is referred to as the *Fourier Transform*.

Fourier Transform

- **Fourier Transform (analysis)**

$$F(\omega) = \mathcal{F}\{f(t)\} = \int_{-\infty}^{\infty} f(t)e^{-j2\pi\omega t} dt$$

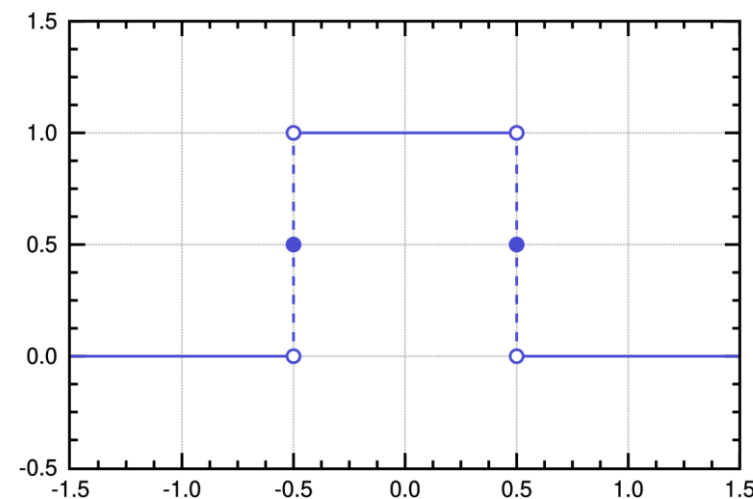
- **Inverse Fourier Transform (synthesis)**

$$f(t) = \mathcal{F}^{-1}\{F(\omega)\} = \int_{-\infty}^{\infty} F(\omega)e^{j2\pi\omega t} d\omega$$

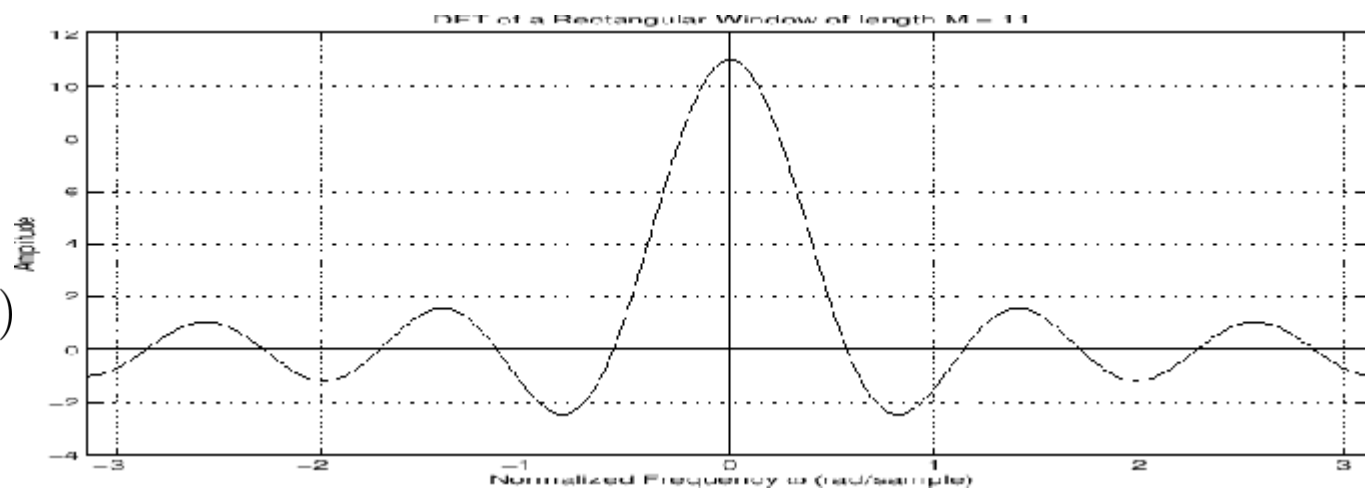
Examples

- **Square function**

$$f(t) = \begin{cases} 1 & -\frac{1}{2} \leq t \leq \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$$



$$\begin{aligned} F(\omega) &= \int_{-\infty}^{\infty} f(t) e^{-j2\pi\omega t} dt = \int_{-\frac{1}{2}}^{\frac{1}{2}} e^{-j2\pi\omega t} dt \\ &= \int_{-\frac{1}{2}}^{\frac{1}{2}} \cos(2\pi\omega t) dt = \frac{\sin(\pi\omega)}{\pi\omega} = \text{sinc}(\omega) \end{aligned}$$



Fourier Transform and Periodic Signals

- While the Fourier Transform is useful for representing/analyzing non-periodic signals, we can generalize it to periodic signals.
- This can be achieved using distributions (i.e., the delta distribution or unit impulse):

$$\delta(x) = 0 \quad \forall x \neq 0$$

- with

$$\int \delta(x) dx = 1$$

Fourier Transform and Periodic Signals

- If we have a unit impulse in the frequency domain:

$$\mathcal{F}\{\delta(\omega - \omega_0)\} = \int_{-\infty}^{\infty} \delta(\omega - \omega_0) e^{j2\pi\omega t} d\omega = e^{j2\pi\omega_0 t}$$

- That is, a unit impulse in the frequency domain corresponds to a conventional (complex) sinusoidal function:

$$\delta(\omega - \omega_0) \xrightarrow{\mathcal{F}} e^{j2\pi\omega_0 t}$$

General Properties

- **Spectrum**

- Given a signal f , we define its spectrum S_f as:

$$S_f(\omega) = |F(\omega)|^2, \quad F(\omega) = \mathcal{F}\{f(t)\}$$

- **Bandwidth**

- Given a signal f we can define the bandwidth, B , as:

$$F(\omega) = 0 \quad \forall \omega \notin B$$

General Properties

Theorem	$f(x)$	$F(k)$
Similarity	$f(ax)$	$\frac{1}{ a } F\left(\frac{k}{a}\right)$
Addition	$f(x) + g(x)$	$F(k) + G(k)$
Homogeneity	$af(x)$	$aF(k)$
Shift	$f(x - a)$	$e^{-i2\pi ak} F(k)$

Convolutional Theorem

- The Fourier Transform converts *convolutions into multiplications*:

$$\mathcal{F}\{f * g\} = \mathcal{F}\{f\}\mathcal{F}\{g\}$$

- Proof:

$$\begin{aligned}\mathcal{F}\{f * g\} &= \int (f * g)(x) e^{-j2\pi\omega x} dx = \int \left(\int f(t) g(x - t) dt \right) e^{-j2\pi\omega x} dx \\ &= \int f \left(\int g(x - t) e^{-j2\pi\omega x} dx \right) dt = \int f(t) (e^{-j2\pi\omega t}) G(k) dt \\ &= G(k) \int f(t) e^{-j2\pi\omega t} dt = F(\omega) G(\omega)\end{aligned}$$

Example: Triangular window

- **Triangular window:**

$$\Lambda(t) = \max(1 - |t|, 0) = \begin{cases} 1 - |t| & |t| < 1 \\ 0 & \text{otherwise} \end{cases}$$

- Alternatively: $\Lambda(t) = \Pi(t) * \Pi(t)$

Rectangular window



- That is, with the convolution theorem, we get

$$\begin{aligned} \mathcal{F}\{\Lambda(t)\} &= \mathcal{F}\{\Pi(t) * \Pi(t)\} = \mathcal{F}\{\Pi(t)\}\mathcal{F}\{\Pi(t)\} \\ &= \text{sinc}(t)\text{sinc}(t) = \text{sinc}^2(t) \end{aligned}$$

Differentiation

- If a function $f(t)$ is *differentiable*, and both f and f' are *integrable*:

$$\mathcal{F}\{f'(t)\}(\omega) = j2\pi\omega F(\omega)$$

- **Proof:**

$$f(t) = \int F(\omega) e^{j2\pi\omega t} d\omega$$

$$f'(t) = \frac{d}{dt} \left(\int F(\omega) e^{j2\pi\omega t} d\omega \right) = \int \underbrace{j2\pi\omega F(\omega)}_{\mathcal{F}\{f'(t)\}} d\omega$$

Energy and Power Theorems

- **Rayleigh's Theorem (Energy)**

$$\int |f(t)|^2 dt = \int |F(\omega)|^2 d\omega$$

- **Parseval's Theorem (Power)**

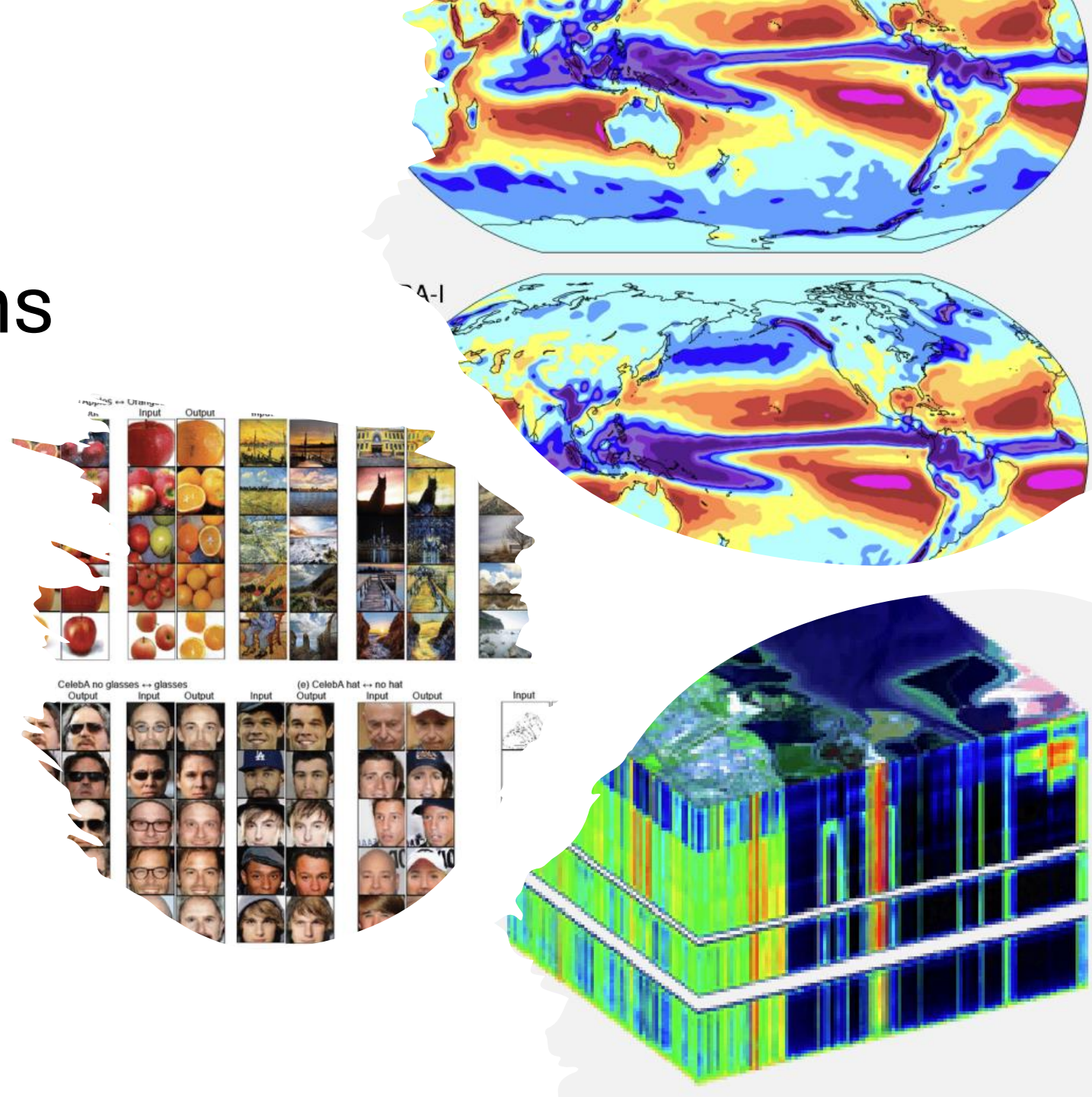
$$\int f(t)g^*(t)dt = \int F(\omega)G^*(\omega)d\omega$$

Summary

Theorem	$f(x)$	$F(k)$
Similarity	$f(ax)$	$\frac{1}{ a } F\left(\frac{k}{a}\right)$
Addition	$f(x) + g(x)$	$F(k) + G(k)$
Homogeneity	$af(x)$	$aF(k)$
Shift	$f(x - a)$	$e^{-i2\pi ak} F(k)$
Convolution	$f(x) * g(x)$	$F(k)G(k)$
Derivative	$f'(x)$	$i2\pi kF(k)$
Rayleigh	$\int_{-\infty}^{+\infty} f(x) ^2 dx = \int_{-\infty}^{+\infty} F(k) ^2 dk$	
Parseval	$\int_{-\infty}^{+\infty} f(x)\overline{g}(x)dx = \int_{-\infty}^{+\infty} F(k)\overline{G}(k)dk$	

Multiple Dimensions

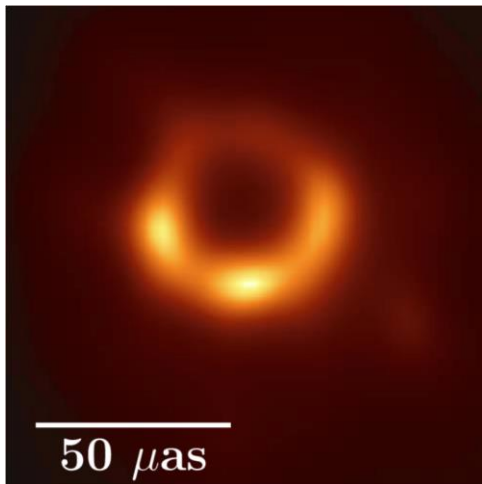
- Until now we focused on one-dimensional signals.
- In nature, we can find a wide range of signals, depending on two or more variables, i.e., lying in a multidimensional space.



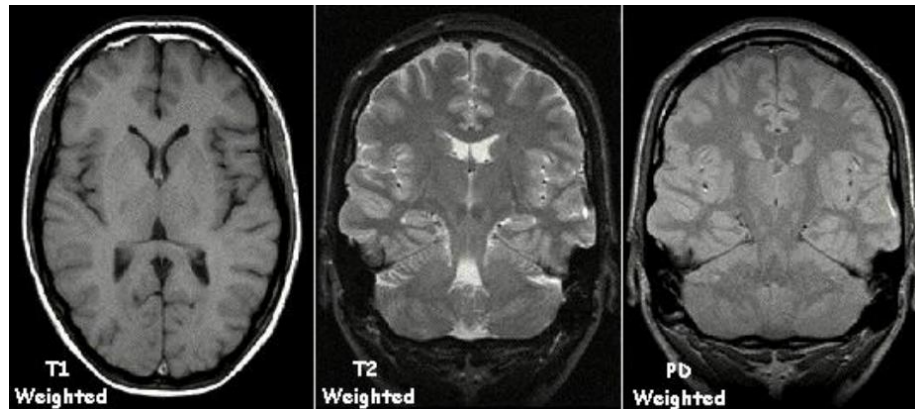
Multidimensional Fourier Domain

- The multidimensional Fourier frequency domain can be used for analysis and imaging.
- **Examples**

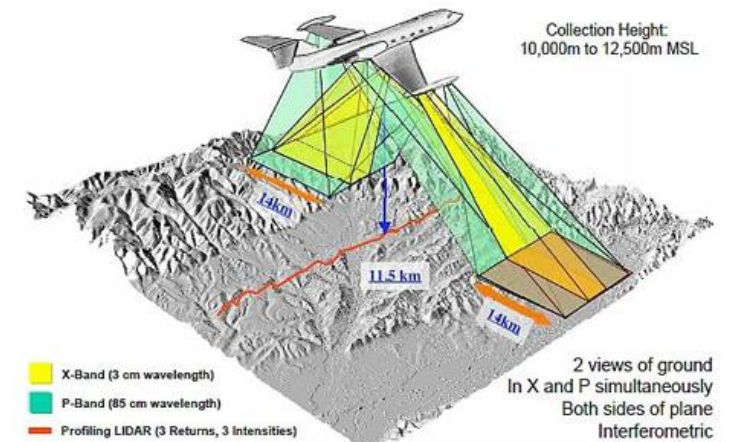
Astronomy



Medical imaging

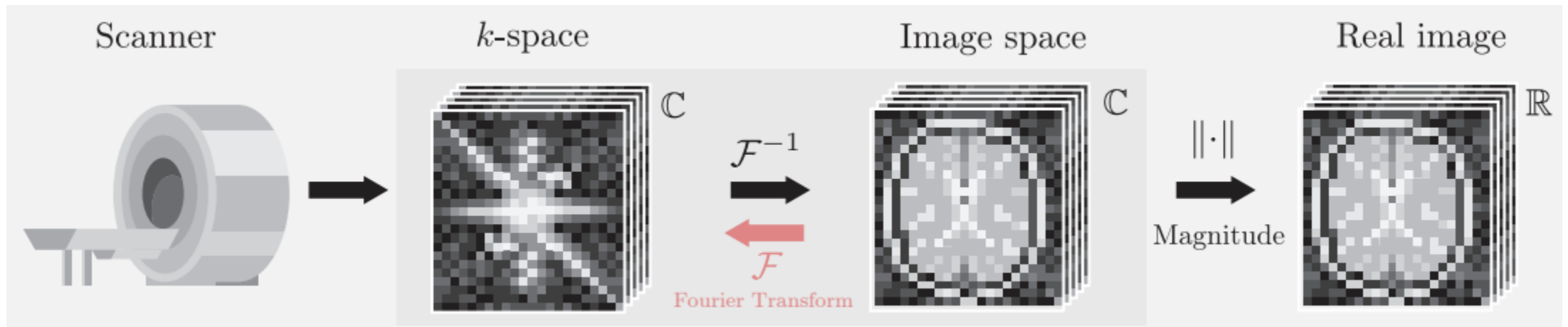


Synthetic Aperture Radar



Multidimensional Fourier Domain

- In contrast to conventional imaging (intensity measured pixel-by-pixel), MRI scanners measures variation in response to different frequencies.
- The result is a complex image in a two-dimensional space of spatial frequencies (k -space).



Multidimensional Fourier Transform

- Consider a signal $f(\mathbf{x})$ with $\mathbf{x} = [x_1, x_2, \dots, x_N]^T$, (x_n is real). Then,

- **Fourier Transform (analysis)**

$$\begin{aligned} F(\boldsymbol{\omega}) &= \mathcal{F}\{f(\mathbf{x})\} = \int \cdots \int f(x_1, \dots, x_N) e^{-j2\pi(x_1\omega_1 + \cdots + x_N\omega_N)} dx_1 \cdots dx_N \\ &= \int_{\mathbb{R}^N} f(\mathbf{x}) e^{-j2\pi\mathbf{x}^T \boldsymbol{\omega}} d\mathbf{x} \end{aligned}$$

- **Inverse Fourier Transform (synthesis)**

$$f(\mathbf{x}) = \mathcal{F}^{-1}\{F(\boldsymbol{\omega})\} = \int_{\mathbb{R}^N} F(\boldsymbol{\omega}) e^{j2\pi\mathbf{x}^T \boldsymbol{\omega}} d\boldsymbol{\omega}$$

- where

$$\boldsymbol{\omega} = [\omega_1 \quad \cdots \quad \omega_N]^T \quad \text{(Multidimensional frequency vector)}$$

Vector of Frequencies

- Let examine a two-dimensional case:

$$e^{\pm 2\pi \mathbf{x}^T \boldsymbol{\omega}} = e^{\pm j 2\pi (x_1 \omega_1 + x_2 \omega_2)}$$

- Regardless of the sign, the exponential function equals 1 if the vector product is an integer, n

$$x_1 \omega_1 + x_2 \omega_2 = n$$

- For a fixed frequency vector, the equation corresponds to a *family of parallel lines* in the (x_1, x_2) plane, normal to the frequency vector
- *Distance* between two consecutive lines:

$$\text{distance} = \frac{1}{\sqrt{\omega_1^2 + \omega_2^2}} = \frac{1}{\|\boldsymbol{\omega}\|}$$

Harmonics and Periodicity

- We may now parametrize $\mathbf{x}$ using the normalized frequency vector ($\mathbf{a} = [a_1, a_2]^T$) is a starting point)

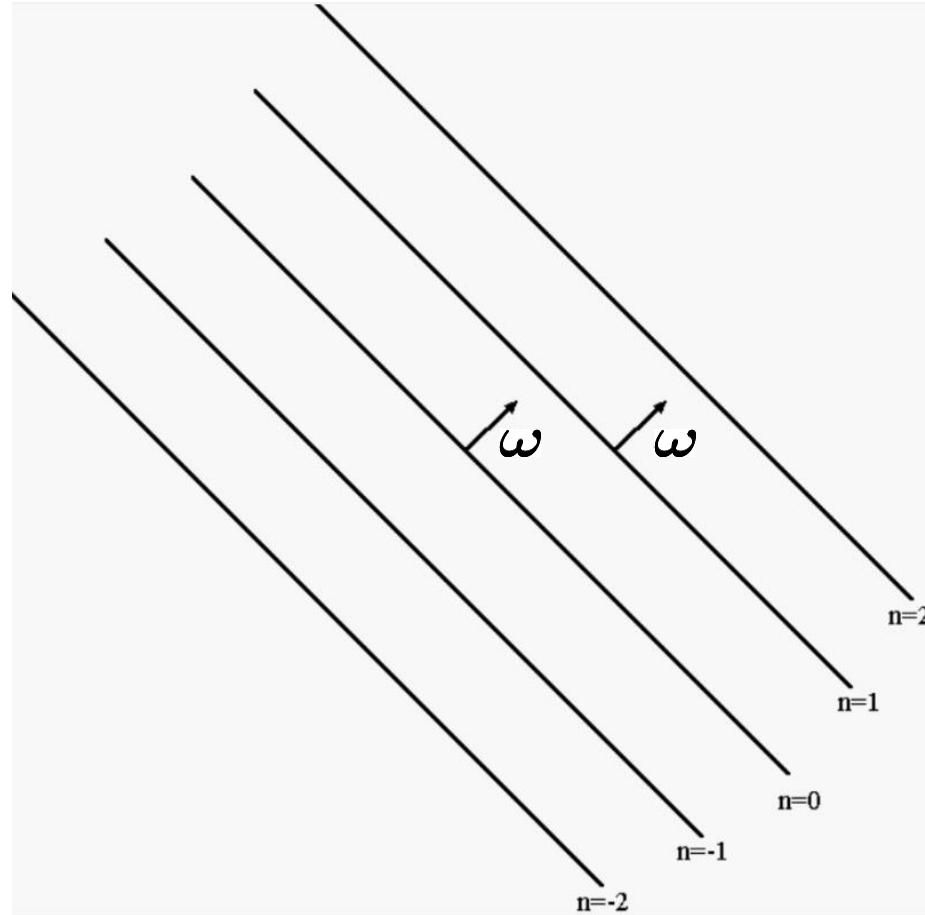
$$\mathbf{x}(t) = \mathbf{a} + t \frac{\boldsymbol{\omega}}{\|\boldsymbol{\omega}\|}$$

- The complex exponential function then becomes

$$e^{\pm j 2\pi \mathbf{x}^T(t) \boldsymbol{\omega}} = e^{\pm j 2\pi \mathbf{a}^T \boldsymbol{\omega}} e^{\pm j 2\pi t \|\boldsymbol{\omega}\|}$$

- The first term is *constant*, not depending on t .
- The second term is *periodic* with period $1/\|\boldsymbol{\omega}\|$.
- Allows us to keep most of the terminology from the 1-D case.

Harmonics and Periodicity



Separable Functions

- In some cases, a function $f(\mathbf{x})$ of N variables may be written as N functions of one variable:

$$f(x_1, \dots, x_N) = f_1(x_1)f_2(x_2) \cdots f_N(x_N)$$

- This significantly simplifies the Fourier analysis, since (in 2-D)

$$\begin{aligned}\mathcal{F}\{f(x_1, x_2)\} &= \int \int e^{-j2\pi(x_1\omega_1 + x_2\omega_2)} f(x_1, x_2) dx_1 dx_2 \\ &= \int \int e^{-j2\pi x_1 \omega_1} e^{-j2\pi x_2 \omega_2} f_1(x_1) f_2(x_2) dx_1 dx_2 \\ &= \mathcal{F}\{f_1(x_1)\} \mathcal{F}\{f_2(x_2)\}\end{aligned}$$

- **In general**

$$\mathcal{F}\{f(x_1, x_2, \dots, x_N)\} = \mathcal{F}\{f_1(x_1)\} \mathcal{F}\{f_2(x_2)\} \cdots \mathcal{F}\{f_N(x_N)\}$$

General Properties

Theorem	$f(\mathbf{x})$	$F(\boldsymbol{\xi})$
Similarity	$f(\mathbf{a}\mathbf{x})$ with $\mathbf{a} \in \mathbb{R}^n$	$\frac{1}{ a_1 a_2 \cdots a_n } F\left(\frac{\xi_1}{a_1}, \frac{\xi_2}{a_2}, \dots, \frac{\xi_n}{a_n}\right)$
Addition	$f(\mathbf{x}) + g(\mathbf{x})$	$F(\boldsymbol{\xi}) + G(\boldsymbol{\xi})$
Shift	$f(\mathbf{x} - \mathbf{a})$	$e^{-i2\pi\mathbf{a}\boldsymbol{\xi}} F(\boldsymbol{\xi})$
Convolution	$f(\mathbf{x}) * g(\mathbf{x})$	$F(\boldsymbol{\xi})G(\boldsymbol{\xi})$
Derivative	$\frac{\partial}{\partial x_i} f(\mathbf{x})$	$i2\pi\xi_i F(\boldsymbol{\xi})$
Rayleigh	$\int_{\mathbb{R}^n} f(\mathbf{x}) ^2 dx = \int_{\mathbb{R}^n} F(\boldsymbol{\xi}) ^2 dk$	
Parseval	$\int_{\mathbb{R}^n} f(\mathbf{x}) \overline{g(\mathbf{x})} dx = \int_{\mathbb{R}^n} F(\boldsymbol{\xi}) \overline{G(\boldsymbol{\xi})} dk$	

General Stretch Theorem

- For any linear transformation, $\mathbf{A}$, of the input data, its corresponding Fourier transform is given by:

$$\mathcal{F}\{f(\mathbf{A}\mathbf{x})\} = \frac{1}{|\det(\mathbf{A})|} F(\mathbf{A}^{-T}\boldsymbol{\omega})$$

- Rotations is a particular kind of linear transformation.
- The rotation matrix, $\mathbf{R}$, has the special property that $\mathbf{R} = \mathbf{R}^{-T}$ and $\det(\mathbf{R}) = 1$, i.e.,

$$\mathcal{F}\{f(\mathbf{R}\mathbf{x})\} = F(\mathbf{R}\boldsymbol{\omega})$$

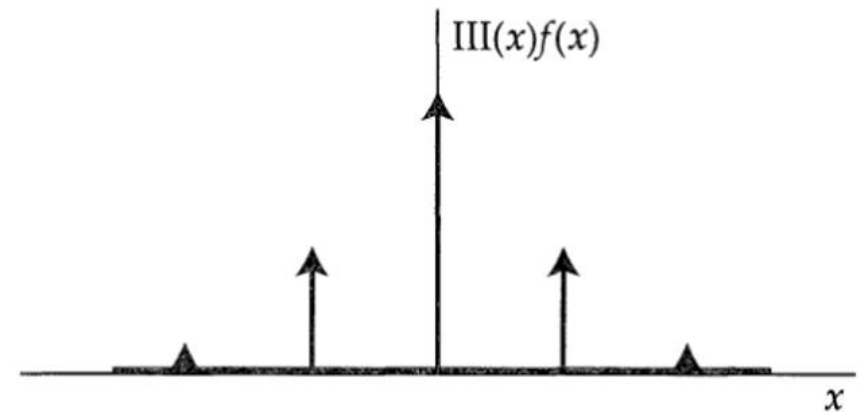
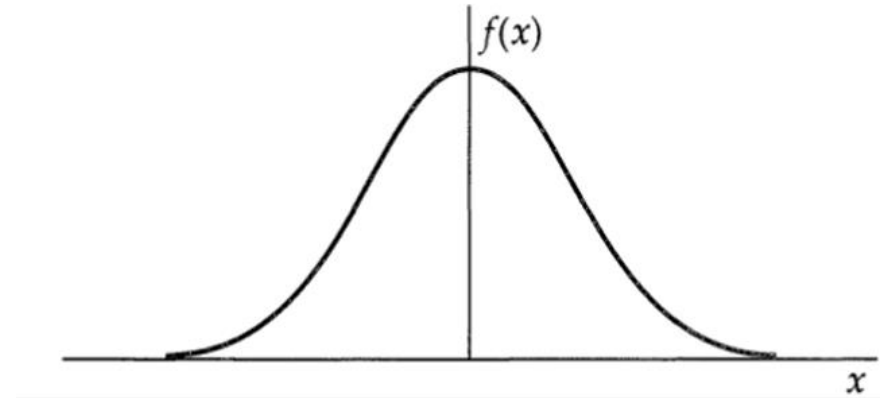
Sampling

- We can define the sampling function as

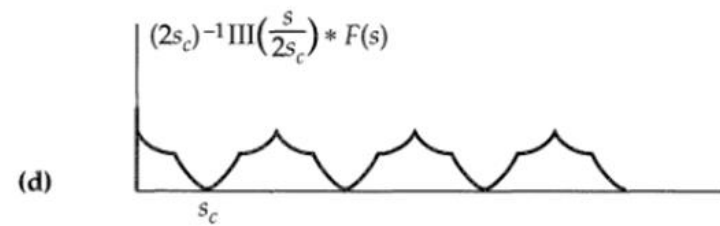
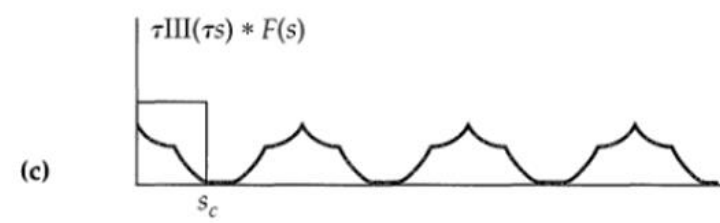
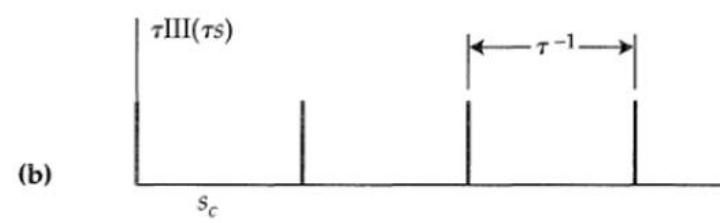
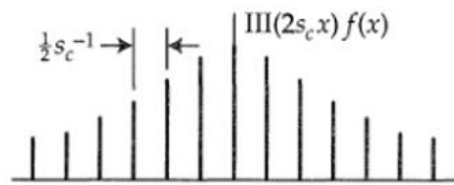
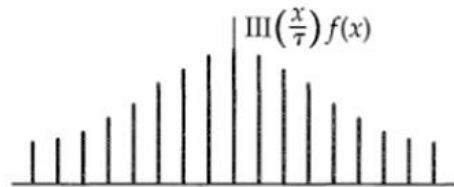
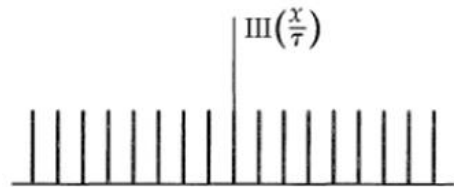
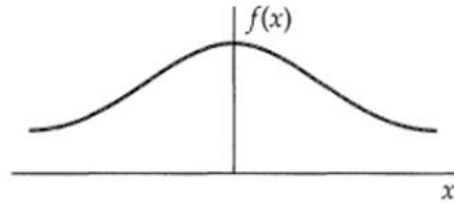
$$\text{III}_T(x) = \sum_{n=-\infty}^{\infty} \delta(x - nT)$$

- Using this function, we can sampling any other function as

$$\text{III}(x)f(x) = \sum_{n=-\infty}^{\infty} f(x - n)$$



Sampling Rate



Nyquist Sampling

- Suppose we are given samples of a function $\rightarrow$ Can we reconstruct the function *perfectly*?
- **Nyquist Sampling Theorem**
 - Let f be a signal with band-limit B .
 - Then, we can reconstruct f *exactly* as long as the sampling frequency is greater than $2B$.

Multidimensional Discrete Fourier Transform

- The MD DFT is a sampled version of the discrete-domain FT.
- Obtained by evaluating it at uniformly spaced sample frequencies:

$$F(\mathbf{k}) = \sum_{n_1=0}^{N_1-1} \cdots \sum_{n_m=0}^{N_m-1} f(\mathbf{x}) e^{-j2\pi \left(\frac{n_1 K_1}{N_1} + \cdots + \frac{n_m K_m}{N_m} \right)}$$

- The inverse MD DFT:

$$f(\mathbf{x}) = \frac{1}{N_1 \cdots N_m} \sum_{n_1=0}^{N_1-1} \cdots \sum_{n_m=0}^{N_m-1} F(\mathbf{k}) e^{j2\pi \left(\frac{n_1 K_1}{N_1} + \cdots + \frac{n_m K_m}{N_m} \right)}$$